

Varignon's theorem of moments

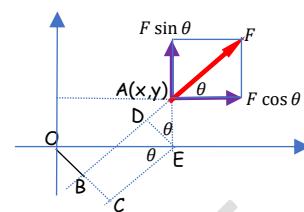
Varignon's theorem states that moment of a resultant force about a moment centre is equal to the algebraic sum of the moments of its components about the same point.

Consider a force F acting at point $A(x, y)$ at an angle θ . The moment of F about a perpendicular axis passing through point O ,

$$M_o = +F \times OB$$

Draw lines $EC \parallel AB$, $OC \perp EC$, and $ED \perp AB$. $\angle AED = \theta$, $\angle OEC = \theta$

$$M_o = F(OC - BC) = F(OC - ED)$$



From $\triangle OCE \gg \sin \theta = \frac{OC}{OE} \gg \gg OC = OE \sin \theta \gg \gg OC = x \sin \theta$

From $\triangle ADE \gg \cos \theta = \frac{ED}{AE} \gg \gg ED = AE \cos \theta \gg \gg ED = y \cos \theta$

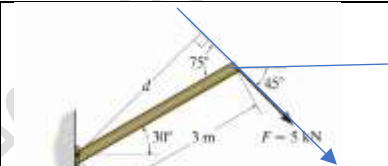
$$M_o = F(OC - ED) = F(x \sin \theta - y \cos \theta) = (F \sin \theta) x - (F \cos \theta) y = (F_y) \cdot x - (F_x) \cdot y$$

$$M_o = \text{moment of } F_y \text{ about } O + \text{moment of } F_x \text{ about } O$$

Moment of resultant force F about O = Algebraic sum of moments of its components about O .

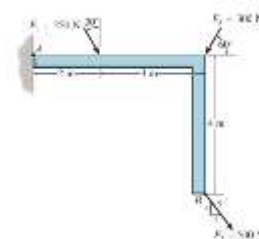
Exercise 1

Determine the moment of F about O .

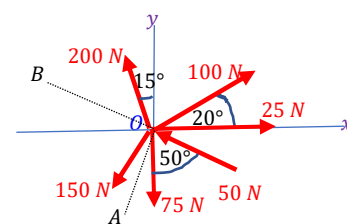


Exercise 2

Find the net moment of all forces about A



Consider a system of concurrent forces acting at O . Let R be the resultant of the force system. If the moment of this resultant force about A (ΣM_A), when calculated, is equal to zero, it implies that either R is zero or the line action of R passes through A . If the moment of these forces about another point B (ΣM_B), which is not in line with OA , is also zero, it again implies that either R is zero or the line action of R passes through B .

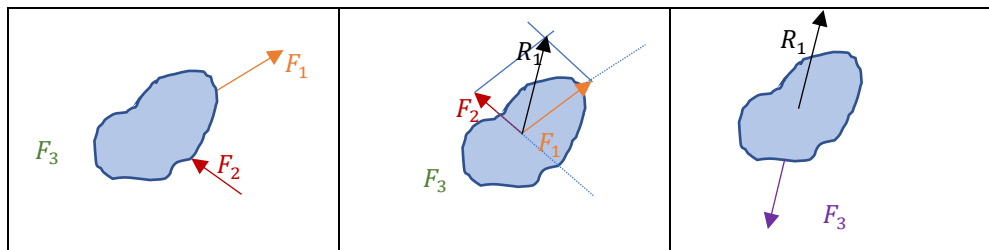


The resultant cannot have two lines of action; hence the resultant R will be zero.

Therefore, if the algebraic sum of the moments of the concurrent forces acting at O about two moment centres A and B are zero (O, A, B are not in a line), then their resultant force is zero. That is, the conditions for equilibrium of concurrent forces are $\Sigma M_A = 0$ and $\Sigma M_B = 0$. We can use this condition also for equilibrium of forces.

Theorem of Three forces

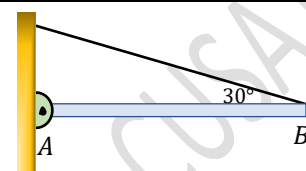
Three forces can be in equilibrium if and only if they are concurrent.



This does not mean that three forces which are concurrent will always be in equilibrium.

Example 1

A beam AB of weight $W = 100\text{ N}$ and length 4 m is hinged at A and supported by a cable at B . Determine the support reaction at A and tension in the cable.



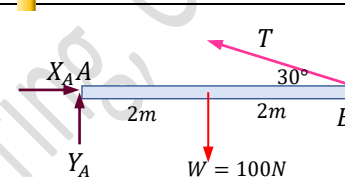
Consider the free body diagram of the beam.

$$\Sigma F_x = 0 \gg X_A - T \cos 30 = 0$$

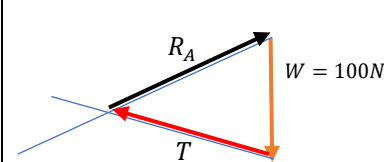
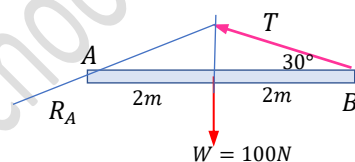
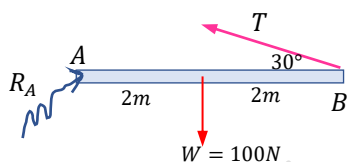
$$\Sigma F_y = 0 \gg Y_A + T \sin 30 - 100 = 0$$

$$\Sigma M_A = 0 \gg T \sin 30 \times 4 - 100 \times 2 = 0 \gg T = 100\text{ N}$$

$$X_A = +86.6\text{ N} \quad Y_A = +50\text{ N}$$

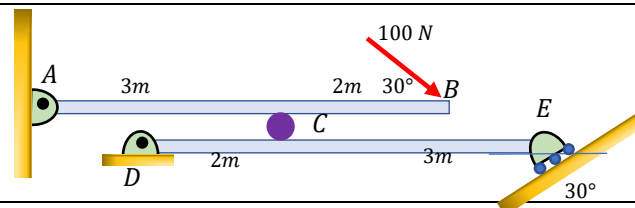


We can also solve this problem graphically. We know that the beam is acted upon by three forces; W , T , and R_A . For equilibrium, these forces must be concurrent. Taking a suitable scale, draw the beam and mark the forces. The reaction R_A should act in such a direction that it will be concurrent with the other two. Then, draw the force triangle using a suitable scale. Draw W to scale. From the end points of force vector W , draw lines parallel to the lines of action of T and R_A to get a triangle. Mark the directions of T and R_A in such a way that it forms a closed triangle.



Example 2

Two beams AB and DE are loaded and supported as shown. Determine the support reactions at A , D , and E .



Consider the equilibrium of beam AB .

$$\Sigma M_A = 0 \gg 100 \sin 30 \times 5 - R_C \times 3 = 0$$

$$R_C = 83.3\text{ N}$$

$$\Sigma F_x = 0 \gg X_A + 100 \cos 30 = 0 \gg X_A = -86.6\text{ N}$$

$$\Sigma F_y = 0 \gg Y_A + R_C - 100 \sin 30 = 0 \gg Y_A = -33.3\text{ N}$$

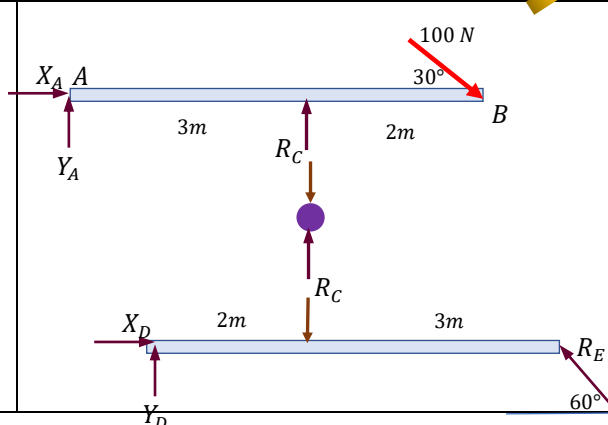
Consider the equilibrium of beam DE .

$$\Sigma M_D = 0 \gg R_E \sin 60 \times 5 - 83.3 \times 2 = 0$$

$$R_E = 38.5\text{ N}$$

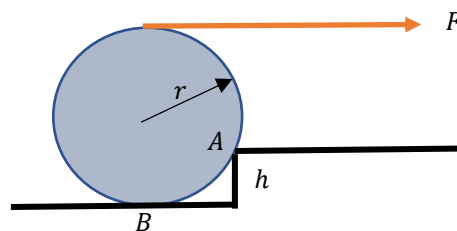
$$\Sigma F_x = 0 \gg X_D - R_E \cos 60 = 0 \gg X_D = 19.5\text{ N}$$

$$\Sigma F_y = 0 \gg Y_D - R_C + R_E \sin 60 = 0 \gg Y_D = 50\text{ N}$$

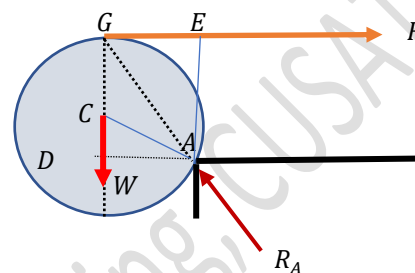


Example 3

A cylinder of radius $r = 12\text{ cm}$ and weight $W = 500\text{ N}$ is to be pulled over a small step of height $h = 6\text{ cm}$. A string is wound around the cylinder and a horizontal force F is applied to the string. What is the force F just sufficient to pull the cylinder over the step?



When the cylinder is just about to be pulled over the step, there will not be any contact between the cylinder and the support at B. Hence, there will not be any support reaction at B. In the free body diagram of the cylinder, there will be only three forces, F , W , and R_A . Therefore, for equilibrium, these three forces should be concurrent.

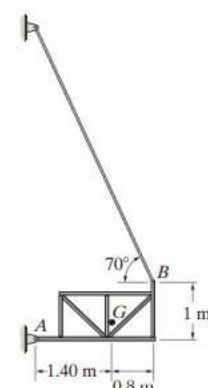


$$\begin{aligned}\Sigma M_A = 0 &\gg W \times AD - F \times AE = 0 \\ 500\text{ N} \times 10.4\text{ cm} - F \times 18\text{ cm} &= 0 \\ F &= 288.9\text{ N}\end{aligned}$$

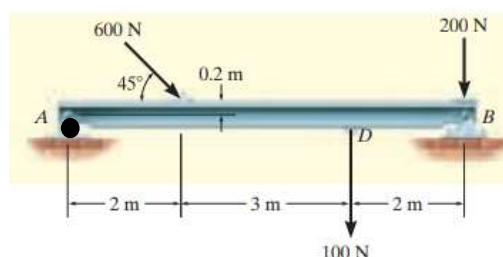
$$\begin{aligned}CD &= r - h = 6\text{ cm} \\ AD &= \sqrt{AC^2 - CD^2} = \sqrt{12^2 - 6^2} = 10.4\text{ cm} \\ AE &= CD + r = 12 + 6 = 18\text{ cm}\end{aligned}$$

Exercise 1

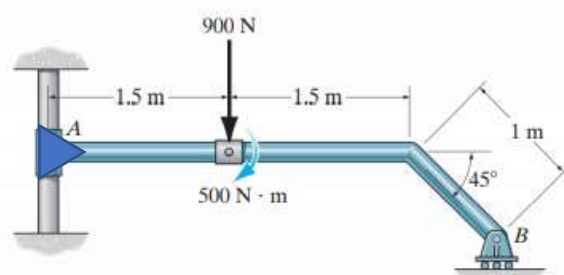
Draw the free body diagram of the oil rig platform shown if the weight of the platform is 2000 N . Determine the tension in the cable and reaction at the support.

**Exercise 2**

A beam AB is loaded and supported as shown. Determine the support reactions.

**Exercise 3**

A beam AB is loaded and supported as shown. Determine the support reactions at the hinged end A and roller support B.



Reference

1. Engineering Mechanics: Timoshenko and Young, McGraw Hill
2. Mechanics for Engineers: Beer & Johnston, McGraw Hill
3. Engineering Mechanics: Merriam & Kraig, Wiley
4. Engineering Mechanics: Hibbeler, Pearson

Prof. Biju N., School of Engineering, CUSAT